# A Comprehensive Literature Survey on Deep Learning Based Methods for Parkinson's Disease Detection Using Dynamic Features of Speech

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**Abstract**

This report presents a comprehensive literature survey on deep learning methods for Parkinson's Disease (PD) detection using dynamic features of speech. The survey covers fundamental concepts in PD speech analysis, including clinical assessment scales, speech processing terminology, validation methodologies, and both traditional machine learning and deep learning approaches. Special emphasis is placed on dynamic articulation features and Bidirectional Long Short-Term Memory (BiLSTM) networks as a novel approach to capturing temporal interdependencies in speech signals. The report provides detailed explanations of clinical metrics (MDS-UPDRS, Hoehn & Yahr scale), speech features (LPC, LAR, prosody, perturbation measures), validation techniques (internal/external validation, speaker splitting, aggregation), and model architectures (Random Forest, SVM, BiLSTM). Additionally, it covers statistical validation concepts, the importance of calibration versus discrimination, and regulatory considerations for clinical deployment. The survey aims to bridge the gap between clinical requirements and computational methods, providing researchers with a comprehensive reference for PD detection from speech signals.

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# 1 Introduction to Parkinson's Disease and Speech Analysis

## 1.1 Background

Parkinson's Disease (PD) is currently the second most frequent neurodegenerative disease, after Alzheimer's disease. The disease presents with two main categories of symptoms:

- **Motor Symptoms**: Tremor, bradykinesia (slowness of movement), rigidity (stiffness), and postural instability (poor balance). These are considered the hallmark features of the disease.
- **Non-Motor Symptoms**: Mood disorders, cognitive dysfunction, pain, sensory dysfunction, and dysautonomia. These are common but often underrecognized in clinical practice.

**Speech Impairment in PD:**

- Motor speech disorders are common among PD patients, occurring in more than half of patients.
- Speech disturbances include very quiet speech (hypophonia), hurried speech (tachylalia), and reduced articulation precision.
- Speech signals offer a non-invasive identification and prediction technology for early detection.
- Detection of voice changes enables early detection and intervention before disabling physical symptoms appear.

**Modern Speech Processing Technology:**
The development of modern speech processing technology relies heavily on interdisciplinary research in:

- Multimodal signal processing
- Artificial Intelligence (AI)
- Human-Computer Interaction (HCI)

**Established Acoustic Correlates of PD:**

Table 1: Acoustic Correlates of Parkinson's Disease

| **Measurement** | **PD Effect** |
|---|---|
| Maximum Phonation Time | Shorter in PD patients |
| Jitter | Higher in PD patients |
| Shimmer | Higher in PD patients |
| Pitch Range | Decreased in PD patients |
| Phonation Threshold Pressure | Increased in PD patients |

## 1.2 Limitations of Previous Approaches

### 1.2.1 Machine Learning Approaches

Most machine learning methods for PD detection fall into two broad categories:

Table 2: Categories of Machine Learning Approaches for PD Detection

| Category | Approach | Limitation |
|---|---|---|
| Traditional ML | Extract static features (MFCC, pitch, jitter, shimmer) + classifier | Ignores temporal dynamics; relies on feature engineering |
| Deep Learning (DL) | Hierarchical feature extraction | Many studies ignore interdependencies in sequential features |

**Dominant Traditional ML Methods:**

- Support Vector Machine (SVM) [12]-[14]
- K-Nearest Neighbor (KNN) [12]
- Naive Bayes (NB) [12], [15], [16]
- Decision Tree [12]
- Genetic Algorithm [14]
- Ensemble Methods [17], [18]

**Traditional ML Pipeline:**

1. Extract global static features
2. Feature selection (LASSO, Relief, LLBFS, etc.)
3. Dimensionality reduction (PCA)
4. Classification

**Deep Learning for PD Detection:**

Table 3: Deep Learning Approaches for PD Detection

| DL Method | Studies | Focus |
|---|---|---|
| CNN | [19]-[22] | Feature-level and model-level combination; spectrogram analysis |
| End-to-End CNN | [20], [21] | Direct audio classification; time-frequency representations |

### 1.2.2 The Gap: Dynamic Features

**The Problem with Static Features:**

Table 4: Limitations of Static Feature Approaches

| Issue | Explanation |
|---|---|
| Temporal Independence | Static features collapse measurements across time, losing the temporal structure of speech |
| Loss of Sequential Information | Interdependencies in feature sequences are ignored, which may contain important diagnostic information |
| Articulation Dynamics | PD affects speech articulation transitions, which are inherently temporal |
| Clinical Relevance | Speech is a dynamic process; static features may miss important patterns that could aid in early detection |

**Prior Evidence of Articulation Impairment in PD:**

- Skodda & Visser (2011): Vowel Articulation Index (VAI) reduced in PD
- Rusz et al. (2013): Monologue most suitable for early PD detection; up to 80% accuracy
- Arias-Vergara et al. (2017): Continuous speech features (prosody, intelligibility, articulation) improve discrimination

**The Observation:** When comparing articulation transitions between Healthy Control (HC) speakers and PD patients:

- The **number of articulation transitions** is significantly different
- The **trend of the fundamental frequency curve** differs
- Paired t-test: $p = 0.042$ ($< 0.05$) — the difference did not occur by chance

# 2 Clinical Assessment Scales for Parkinson's Disease

## 2.1 MDS-UPDRS

MDS-UPDRS stands for Movement Disorder Society–Unified Parkinson's Disease Rating Scale. It is a clinical scale used to measure the severity and progression of Parkinson's disease (PD). It evaluates several aspects, including:

- **Motor symptoms** - Physical manifestations of PD
- **Non-motor symptoms** - Cognitive, mood, and autonomic changes
- **Daily activities** - Impact on activities of daily living
- **Motor complications** - Treatment-related issues

The score is generally higher when Parkinson's symptoms are more severe. For a speech-processing paper, MDS-UPDRS may be used as the clinical ground-truth/reference measure against which speech-derived features or model predictions are compared.

## 2.2 Hoehn and Yahr (H&Y) Scale

The H&Y scale measures the stage/severity of Parkinson's disease, mainly based on motor symptoms and functional impairment. Unlike MDS-UPDRS, which gives a more detailed severity score, H&Y gives a disease stage.

Table 5: Hoehn and Yahr Scale Stages

| Stage | Meaning |
|---|---|
| 1 | Symptoms mainly on one side |
| 2 | Symptoms on both sides, but balance generally preserved |
| 3 | Postural instability appears, but person is still physically independent |
| 4 | Severe disability; needs considerable assistance |
| 5 | Very severe; generally requires a wheelchair/bed unless aided |

## 2.3 Time After Diagnosis (TAD)

TAD simply means how much time has passed since the person was diagnosed with Parkinson's disease. This can be important because speech characteristics may change as PD progresses.

**Example:**

- Diagnosis: January 2022
- Speech recording: January 2025
- TAD = 3 years

# 3 Speech Processing Fundamentals

## 3.1 Prosody

In speech processing, prosody refers to the aspects of speech that go beyond the individual speech sounds (phonemes). Think of it as the rhythm, melody, and timing of speech.

**Main Components of Prosody:**

Table 6: Components of Prosody

| **Component** | **Definition** | **Example** |
|---|---|---|
| Pitch (F0) | How high or low the voice sounds; related to fundamental frequency of vocal-fold vibration | Pitch rising at the end of a question |
| Duration | How long sounds, syllables, or pauses last | Unusually long pauses in speech |
| Intensity/Loudness | How strongly or loudly something is produced; measured using acoustic energy/amplitude, often in dB | Reduced loudness in PD |
| Speech Rate | How quickly someone produces speech; measured in syllables per second or words per minute | Slower rate in PD |
| Pauses | Silent intervals between words or phrases; number and duration can be important features | Increased pause frequency in PD |
| Rhythm | Pattern of stressed/unstressed syllables and timing across speech | Reduced rhythmic variation in PD |

**In Parkinson's Disease:** Prosody is particularly relevant because people with Parkinson's disease can show reduced or abnormal prosodic variation. For example, speech may become:

- More monotonous → reduced pitch variation
- Quieter → reduced intensity
- Slower or less rhythmically regular
- Characterized by abnormal pauses
- Less expressive overall

A speech-processing model might extract features such as:

$$\text{Prosodic Features} = \{F0, \text{Intensity}, \text{Duration}, \text{Speech Rate}, \text{Pause Characteristics}\} \quad (1)$$

## 3.2 Linear Predictive Coding (LPC)

LPC is a speech-processing technique that represents the spectral characteristics of speech using a small number of coefficients. The basic idea is to predict the current speech sample from previous speech samples.

$$\hat{s}(n) = a_1 s(n-1) + a_2 s(n-2) + \cdots + a_p s(n-p) \tag{2}$$

where $a_1, a_2, \ldots, a_p$ are the LPC coefficients. These coefficients capture information about the vocal-tract shape and resonances (formant structure).

## 3.3 Log Area Ratio (LAR)

LAR is another way of representing vocal-tract information, closely related to the LPC/reflection coefficients. It describes the relationship between the cross-sectional areas of consecutive sections of the vocal tract.

$$\text{LAR}_i = \log\left(\frac{1+k_i}{1-k_i}\right) \tag{3}$$

where $k_i$ is a reflection coefficient.
**Key Relationship:**

- LPC $\rightarrow$ represents vocal-tract characteristics using prediction coefficients
- LAR $\rightarrow$ represents related vocal-tract characteristics using log area ratios
- LPC and LAR can contain overlapping/correlated information

## 3.4 Perturbation Measures

### 3.4.1 Jitter and PPQ

**Jitter** measures temporal perturbation of fundamental frequency. **PPQ (Pitch Perturbation Quotent)** measures short-term variation in the fundamental frequency (F0) from one vocal period to another. In simpler terms, PPQ measures how much the pitch of a sustained voice signal fluctuates from cycle to cycle.

Higher PPQ indicates greater pitch-period variation, which means a more irregular/unstable voice.

### 3.4.2 Shimmer and APQ

**Shimmer** measures temporal perturbation of amplitude. **APQ (Amplitude Perturbation Quotent)** measures short-term variation in the amplitude/loudness of consecutive vocal cycles. In simple terms, APQ measures how much the loudness/strength of successive vocal cycles fluctuates.

Higher APQ indicates greater amplitude variation, which means a more irregular voice.

Table 7: Comparison of PPQ and APQ

| Feature | Measures Variation In | Related To |
|---|---|---|
| PPQ | Pitch / Frequency | Jitter |
| APQ | Amplitude / Loudness | Shimmer |

## 3.5 Maximum Phonation Time (MPT)

Maximum Phonation Time is the maximum amount of time a person can sustain a vowel sound on one breath. Typically, the person takes a deep breath and sustains something like /a/ for as long as possible.

MPT measures aspects of:

- Breath support
- Vocal-fold function
- Phonatory control

**Example:**

- Person A sustains /a/ for 18 seconds
- Person B sustains /a/ for 10 seconds
- Person B has a shorter maximum phonation time

## 3.6 Vowel Space Measures

### 3.6.1 Vowel Space Area (VSA)

VSA represents the area occupied by vowels in the F1–F2 formant space. Usually, vowels such as /i/, /a/, and /u/ are plotted using their formant frequencies.

**The Vowel Space Concept:** We can represent vowels using their formant frequencies, mainly F1 and F2. Think of it as a map:

- Different vowels occupy different positions in this acoustic space
- If vowels are far apart → they are acoustically distinct
- If vowels move closer together → they become acoustically similar

**Why is this important in PD speech?** A speaker may have reduced articulatory precision, causing different vowels to move toward the center of the vowel space:

- Vowels clearly separated → large vowel space (healthy)
- Vowels clustered together → small vowel space (PD)

This is called **vowel centralization** and is a characteristic finding in PD speech.

### 3.6.2 Vocal Pentagon Area (VPA)

VPA is similar in idea to VSA, but instead of using only three vowels, it uses five vowels to form a polygon/pentagon in the acoustic vowel space. VPA measures the area enclosed by five selected vowels in the F1–F2 space.

### 3.6.3 Formant Centralization Ratio (FCR)

FCR measures how centralized the vowel formants are. In simplified terms, it tells us how much the speaker's vowels have moved toward the center of the vowel space.

- Higher centralization → vowels become acoustically more similar/less distinct
- FCR is often useful when studying vowel articulation changes

## 3.7 Onset and Offset

Table 8: Onset and Offset Definitions

| **Feature** | **Meaning** |
|---|---|
| Onset | Transition from unvoiced to voiced segments; beginning of voicing |
| Offset | Transition from voiced to unvoiced segments; end of voicing |

These onset/offset measures can capture how smoothly or abruptly voicing starts and stops. PD patients often show abnormal unvoiced sounds and have difficulty beginning and/or stopping vocal fold vibration.

# 4 Experimental Datasets and Setup

## 4.1 GYENNO Dataset

Table 9: GYENNO Dataset Characteristics

| **Aspect** | **Detail** |
|---|---|
| Total Subjects | 45 (15 HC, 30 PD) |
| Gender | 25 female, 20 male |
| Age Range | 37-75 years |
| PD Stages | HY (Hoehn & Yahr) stage 1-5 |
| Speech Tasks | Sustained monophonic /a/ ( 5.5s), Short sentence ( 5.5s) |
| Total Samples | 268 |
| Recording Device | Smartphone |
| Sampling Rate | 96 kHz |
| Feature Extraction | NeuroSpeech software |
| Language | Mandarin Chinese (all subjects native speakers) |

## 4.2 Experimental Setups

Researchers use four different datasets/experimental setups, designed to answer different questions:

1. **AllPD**: Contains all available Parkinson's disease speakers (Early PD + non-Early PD). This is the normal/baseline setup where the model learns from PD speakers across different stages of disease.

2. **AllPD-sub**: A smaller subset of AllPD, created so that its speaker distribution is comparable to EarlyPD. This controls for the amount of training data, allowing comparison between EarlyPD and AllPD without the advantage of more speakers.

3. **EarlyPD**: Contains only EarlyPD speakers (early-stage Parkinson's speakers only). This tests whether a model can be trained specifically using early-stage PD data.

4. **EarlyPD + Private**: Takes EarlyPD public dataset plus additional external/private dataset. This tests whether adding external data improves the model's ability to learn early-PD characteristics.

Table 10: Experimental Training Setups

| **Training Setup** | **What Model Sees** |
|---|---|
| AllPD | Early + non-Early PD |
| AllPD-sub | Smaller, distribution-matched AllPD |
| EarlyPD | Early PD only |
| EarlyPD + Private | Early PD + additional external/private data |

# 5 Validation Methodologies

## 5.1 Internal Validation

Internal validation means evaluating how well the model is expected to perform using data from the same overall study/dataset used for model development. Common methods include:

- Cross-validation
- k-fold cross-validation
- Bootstrap validation
- Train/validation splits

**Example:**

1. One dataset
2. Split into folds

3. Train on some folds
4. Validate on remaining fold
5. Repeat

## 5.2 External Validation

External validation means testing a trained model on a completely independent dataset that was not used to train or develop the model. The purpose is to check whether the model works on new data from outside the dataset on which it was developed.

**Example:**

```
Training dataset -> Train model -> Model -> Completely different dataset -> External
```

This gives stronger evidence that the model generalizes rather than simply learning the characteristics of the original dataset.

## 5.3 Key Differences: Internal vs. External Validation

Table 11: Comparison of Internal and External Validation

| | **Internal Validation** | **External Validation** |
|---|---|---|
| Data Source | Same study/dataset | Independent dataset |
| Purpose | Check model development/performance | Check generalization |
| Strength | Useful for model selection | Stronger evidence of real-world generalization |
| Example | 5-fold cross-validation | Test on another hospital/dataset |

If a paper says "internally and externally validated," it means:

1. They first checked whether the model works reliably within their own dataset
2. Then checked whether it still works on an independent dataset

**Important Note:** External validation is not a problem in itself. In fact, it is one of the strongest ways to test whether a model really generalizes. The problem is that external validation is difficult to obtain and can reveal that a model that worked well internally does not work well on new data.

## 5.4 Why External Validation Is Difficult in Speech/PD

Speech data are strongly affected by factors other than the disease:

- Microphone/recording environment
- Language/accent
- Age and sex

- Speaking task
- Disease severity
- Medication state
- Dataset collection protocol

A model might accidentally learn dataset-specific characteristics rather than the actual PD-related speech characteristics.

## 5.5 Speaker Split

Speaker split means dividing the dataset based on different speakers (people) rather than individual speech recordings. This ensures that all recordings from one person stay in the same dataset partition (train/validation/test).

**Example:**

- 70 speakers → training
- 15 speakers → validation
- 15 speakers → test

**Why is this important?** A single speaker may have many speech recordings. If you randomly split recordings, the model may have already heard the same person's voice characteristics during training, making test performance artificially good.

**In one sentence:** Speaker split = ensuring that all recordings from one person stay in the same dataset partition (train/validation/test). This is especially important for PD speech classification, because otherwise the model might learn who the speaker is rather than whether their speech contains characteristics associated with Parkinson's disease.

## 5.6 Aggregation-Based Samples

Aggregation-based samples means that instead of treating each individual speech recording/utterance as one independent sample, multiple recordings from the same speaker are combined (aggregated) to create a sample.

**Example:**

```
Speaker A
  Recording 1
  Recording 2
  Recording 3
  Recording 4
  Recording 5
        ↓
   Aggregation
        ↓
   One speaker-level sample
```

The aggregation could involve taking the mean, median, majority vote, or combining predictions/features from those recordings.

**Why do this?** Because the research question may be about the person/speaker, not an individual recording:

- Recording-level prediction: "Is this particular recording from a person with PD?"
- Aggregation-based prediction: "Based on all/multiple recordings from this person, is this person likely to have PD?"

**Advantages of Aggregation:**

- Reduces recording-level noise
- Gives a more stable estimate of the speaker's characteristics
- Matches the clinical question better
- Prevents over-counting speakers

**Important Caveat:** Aggregation is not automatically better. If individual recordings contain useful information, aggregating too early can lose that information.

# 6 Model Calibration and Discrimination

## 6.1 Discrimination

Discrimination asks: "Can the model distinguish PD from non-PD?"

For example, suppose:

- Patient A $\rightarrow$ PD $\rightarrow$ predicted probability 0.90
- Patient B $\rightarrow$ No PD $\rightarrow$ predicted probability 0.10

That's good discrimination because the model ranks the PD patient higher.

## 6.2 Calibration

Calibration is about whether the probability predicted by the model can actually be trusted. Suppose a model says: PD probability = 80%. A well-calibrated model means that among many cases where the model predicts around 80%, approximately 80% of them actually have PD.

So calibration asks: "Can I trust the probability that the model gives me?"

**Example:** Imagine 100 patients. The model predicts 80% probability of PD for all 100.

- If approximately 80 actually have PD $\rightarrow$ the model is well calibrated
- But if only 50 actually have PD $\rightarrow$ the model is overconfident

## 6.3 Balanced Accuracy vs. Brier Score

**Balanced Accuracy:** Measures classification performance/discrimination. It considers performance on both classes:

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \tag{4}$$

**Brier Score:** Evaluates the quality of probabilistic predictions. For binary classification:

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2 \tag{5}$$

where:

- $p_i$ = predicted probability
- $y_i$ = actual outcome (0 or 1)

Lower Brier score = better probabilistic prediction/calibration.

## 6.4 Key Clinical Finding

"Improvements in balanced accuracy under raw audio input were not consistently accompanied by lower Brier scores."

This means: when the model was given raw audio, sometimes it became better at distinguishing PD from non-PD, but its predicted probabilities did not necessarily become more reliable.

Table 12: Example of Discrimination vs. Calibration

| **Model** | **Balanced Accuracy** | **Brier Score** |
|---|---|---|
| Model A | 75% | 0.15 |
| Model B | 82% ↑ | 0.19 ↑ |

Model B is better at classification, but its Brier score is worse. Therefore, better discrimination does not necessarily mean better calibration.

**Why is this important clinically?** Clinical systems often need more than "PD / No PD" — they may need "There is an estimated 75% probability of PD." That probability needs to be meaningful and trustworthy.

**In one line:**

- Discrimination: Can the model separate PD from non-PD?
- Calibration: When the model says 80% probability, does that really correspond to roughly an 80% chance?

# 7 Machine Learning Models and Algorithms

## 7.1 Random Forest and Breiman Algorithm

Random Forest (RF) is an ensemble learning method introduced by Leo Breiman. The Breiman algorithm works as follows:

1. Create many decision trees
2. For each tree, take a bootstrap sample of the training data (sample with replacement)
3. At each split of a tree, consider only a random subset of features rather than all features
4. Grow the trees according to specified tree parameters
5. Combine the predictions of all trees:
    - Classification: majority vote
    - Regression: average prediction

```
Training data -> Bootstrap samples
         ↓    ↓    ↓    ↓
     Tree1 Tree2 Tree3 ... Tree100
         ↓    ↓    ↓    ↓
         Predictions -> Majority vote -> Final prediction
```

## 7.2 Minimum Terminal Node Size = 5

This is a Random Forest/decision-tree parameter. A terminal node (leaf) is a final node of a decision tree where the tree stops splitting.

**Minimum terminal node size = 5** means:

- A terminal/leaf node must contain at least 5 training samples
- The tree is not allowed to keep splitting until a leaf contains only 1, 2, 3, or 4 samples

**Why do this?**

- Prevents the tree from becoming excessively specific to the training data
- Creates somewhat simpler/smoother trees
- Can reduce overfitting

## 7.3 Bootstrapping in Random Forest

Bootstrapping means creating a new training dataset for each decision tree by randomly sampling the original training data with replacement.

**Example:** Suppose you have 5 training samples: $[A, B, C, D, E]$

For one tree:

- Randomly select 5 samples with replacement: $[A, C, C, E, B]$

For another tree:

- Randomly select 5 samples with replacement: $[D, D, A, E, C]$

For another:

- Randomly select 5 samples with replacement: $[B, E, E, A, D]$

**Why "with replacement"?** After selecting a sample, we put it back before selecting the next one. So C can be selected multiple times, while D might not be selected at all.

Samples not selected for a particular bootstrap sample are called **out-of-bag (OOB)** samples, which can be used to estimate the tree/forest's performance without a separate validation set.

## 7.4 Support Vector Machine (SVM)

### 7.4.1 RBF Kernel

RBF (Radial Basis Function) kernel is commonly used with SVM to handle nonlinear relationships. Instead of trying to separate classes with a straight line/plane, RBF allows a curved decision boundary.

The kernel is:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \tag{6}$$

In simple terms, RBF measures how similar two samples are based on their distance:

- Nearby samples $\rightarrow$ high similarity
- Far-away samples $\rightarrow$ low similarity

### 7.4.2 Regularization Parameter C

For an SVM, $C$ is the regularization parameter. It controls the trade-off between:

- Correctly classifying training samples
- Keeping the decision boundary/generalization reasonable

A larger $C$ penalizes classification errors more strongly, so the model tries harder to classify training data correctly. $C = 10$ means the SVM used 10 as its regularization parameter.

## 7.5 LBFGS Optimizer

LBFGS = Limited-memory Broyden–Fletcher–Goldfarb–Shanno. It is an optimization algorithm used to find the model parameters that minimize a loss function.

In simple terms: LBFGS is the method used to adjust the model's parameters to find a good solution. It is commonly used for models such as logistic regression, especially when the dataset is not extremely large.

## 7.6 Logistic Regression and Multicollinearity

Logistic regression can become problematic when features are highly correlated (multicollinearity). Suppose your features are:

- LPC coefficient 1
- LPC coefficient 2
- LAR feature
- Another feature that is almost a linear combination of LPC coefficients

If two features contain almost the same information, logistic regression has difficulty determining how much weight to give each one.

**What can happen?** Suppose $x_1 =$ LPC feature and $x_2 \approx 0.99x_1$. Both carry nearly the same information.

The model might produce something like:

$$P(PD) = \sigma(10x_1 - 9.8x_2 + 0.5) \tag{7}$$

Another training split might produce:

$$P(PD) = \sigma(3x_1 - 2.8x_2 + 0.5) \tag{8}$$

Both models can make similar predictions, but their individual coefficients are very different.

**Consequences of high correlation:**

- Unstable coefficients
- Large standard errors
- Difficulty interpreting which feature is important
- Sometimes worse generalization

**Common approaches to address multicollinearity:**

- Remove highly correlated features
- PCA $\rightarrow$ convert correlated features into less-correlated components
- Regularization (L2/ridge logistic regression, L1/lasso for feature selection)
- Use a model less sensitive to multicollinearity

**Important Note:** Correlated features do not automatically make logistic regression bad at prediction. If your goal is simply classification, logistic regression can still perform well. The problem is especially serious when you want to interpret the coefficients.

# 8 Statistical Validation

## 8.1 T-Test and Paired T-Test

A t-test is a statistical test used to answer a simple question: "Are these two groups/measurements actually different, or could the difference just be due to random variation?"

**Simple Example:** Suppose you measure a speech feature:

- PD group average = 10
- Healthy group average = 12

There is a difference of 2. But maybe if you took another set of people, the averages would also differ by 2 just because of random variation. The t-test checks how strong the evidence is that the difference is real.

**The Result: p-value** The t-test gives you a p-value. For example, $p = 0.042$. If we use the common significance level $\alpha = 0.05$, then $0.042 < 0.05$, so we say the difference is statistically significant.

**Important Correction:** Don't say: "There is only a 4.2% chance that the difference happened by chance." That's not exactly what the p-value means. Instead say: "If there were actually no difference between the groups, getting a difference this large (or larger) would be relatively unlikely."

**Paired T-Test:** A paired t-test is used when the two measurements are related/matched. For example, measuring the same 44 people before and after treatment, or matched observations from the same subjects.

Thus:

- t-test → Are two means significantly different?
- paired t-test → Are two related/matched means significantly different?

## 8.2 Interdependencies in Sequences of Features

This means that features occurring at different time points are related to each other. For speech, imagine a sequence:

$$f_1, f_2, f_3, f_4, \ldots \tag{9}$$

The value of $f_3$ may depend on or be related to what happened at $f_2$. For example, speech features change gradually rather than randomly.

So "interdependencies in sequences of features" means the model considers the temporal relationship between successive speech features, rather than treating every feature/frame as completely independent.

## 8.3 Dataset-Dependent Gains

This means that an improvement in model performance depends on which dataset or experimental setup is being used.

**Example:**

- Model A improves on Dataset 1
- But the same improvement is not seen on Dataset 2

- Then we say the gain is dataset-dependent

This does not mean "This method always improves performance." It means "This method improves performance under certain datasets/data conditions, but the improvement is not necessarily consistent everywhere."

This can happen because datasets differ in:

- Number of speakers
- Disease severity
- Recording conditions
- Speech tasks
- Demographics
- Class distribution
- Audio quality

# 9 Regulatory Considerations

## 9.1 FDA/EMA Approval

FDA/EMA approval processes mean the regulatory procedures required before a medical product/device can be legally marketed and used in patients. In the context of a research paper, if they say "the method still requires FDA/EMA approval," they mean:

Even if a research model shows good accuracy, it cannot automatically be used as a clinical medical tool. It must go through regulatory evaluation.

## 9.2 FDA (U.S. Food and Drug Administration)

For a medical device, the FDA uses different pathways depending on the device's risk and characteristics, such as 510(k), De Novo, or PMA. Very roughly:

```
Medical device -> Determine risk/classification -> Choose regulatory pathway
-> Provide safety + performance evidence -> FDA review -> Clearance/approval
-> Post-market monitoring
```

For higher-risk Class III devices, the PMA (Premarket Approval) pathway requires scientific evidence providing reasonable assurance that the device is safe and effective.

## 9.3 EMA (European Medicines Agency)

For medical devices, the situation is slightly different from the FDA. Under the EU Medical Devices Regulation, manufacturers generally go through a conformity assessment, often involving an independent notified body. If the requirements are satisfied, the device can receive the CE mark and enter the EU market.

## 9.4 Why Is This Relevant to an AI/Speech Model?

Suppose your model detects Parkinson's disease from speech with 90% accuracy. That doesn't mean it can immediately be used to diagnose patients. Regulators would want evidence concerning:

- Is the system safe?
- Does it work reliably on different populations?
- Is its performance clinically meaningful?
- Has it been properly validated?
- Does it behave consistently in the intended clinical setting?

**Key Takeaway:** Good research performance  automatically approved clinical product. There is a separate regulatory validation process before a medical AI/device can be used commercially or clinically for its intended purpose.

## 9.5 No Formal Meta-Analysis

A meta-analysis is a statistical method where researchers combine the numerical results of multiple studies to obtain an overall estimate. For example, if 10 papers report the effect of a speech feature on PD detection, a meta-analysis might combine their results to calculate an overall effect.

**Why No Meta-Analysis?** Because there was high heterogeneity among the studies — they are too different from each other.

**Example of Heterogeneity:**

Table 13: Heterogeneity in PD Detection Studies

| Study | Dataset | Features | Model |
|---|---|---|---|
| Paper 1 | 50 subjects | LPC | SVM |
| Paper 2 | 200 subjects | MFCC | RF |
| Paper 3 | 80 subjects | Prosody | CNN |
| Paper 4 | 40 subjects | LAR | Logistic Regression |

These studies aren't measuring exactly the same thing under the same conditions. Therefore, combining their numbers into one statistical estimate could be misleading or invalid.

**Therefore:** High heterogeneity prevents quantitative synthesis. Because the studies differ substantially in datasets, features, methods, populations, and evaluation procedures, their numerical results cannot be reliably combined into a single overall statistical result.

Instead, researchers perform a qualitative/narrative synthesis — they discuss and compare the findings without mathematically pooling them.

# 10 Research Objectives and Novel Contributions

## 10.1 Research Questions

1. Do dynamic speech features (articulation transitions) provide better discrimination than static features?

2. Can Bidirectional LSTM models effectively capture time-series dynamic features for PD detection?

3. How do dynamic features compare to traditional ML with static features and end-to-end CNN approaches?

## 10.2 Novel Contributions

Table 14: Novel Contributions of the Research

| **Contribution** | **Significance** |
|---|---|
| Articulation Transition Analysis | Demonstrated significant differences in transition count and F0 trend between HC and PD |
| Dynamic Feature Extraction | Measured energy content in unvoiced→voiced (onset) and voiced→unvoiced (offset) transitions |
| Bidirectional LSTM Application | First application of Bidirectional LSTM to dynamic articulation features for PD detection |
| Comprehensive Comparison | Compared BiLSTM with traditional ML methods and CNN approaches |

## 10.3 Key Hypothesis

PD patients produce abnormal unvoiced sounds and have difficulty beginning and/or stopping vocal fold vibration. This can be observed by modeling the frequency content of unvoiced frames and transitions between voiced and unvoiced sounds.

# 11 Conclusion

This comprehensive literature survey has covered the essential concepts in Parkinson's Disease detection using dynamic speech features. Key takeaways include:

1. **Clinical assessment scales** (MDS-UPDRS, H&Y) provide the ground truth for model development and evaluation.

2. **Dynamic features** capture important temporal interdependencies that static features miss, particularly articulation transitions.

3. **Bidirectional LSTMs** offer a powerful approach to model sequential features, capturing both past and future context.

4. **Proper validation** is critical: speaker splits prevent speaker-dependent leakage, aggregation reduces recording-level noise, and external validation tests generalization.

5. **Calibration matters** alongside discrimination: high accuracy alone is insufficient for clinical decision-making when probabilistic predictions are needed.

6. **Regulatory approval** is required before any AI-based medical tool can be deployed clinically, with FDA/EMA pathways ensuring safety and effectiveness.

7. The research community faces challenges of **heterogeneity** in datasets, features, and methods, making meta-analysis difficult but highlighting the need for standardized evaluation protocols.

The integration of deep learning methods with dynamic speech features represents a promising direction for early, non-invasive PD detection, with the potential to enable timely intervention and improved quality of life for patients.

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